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ASTR rent fee deductions

Updated 2 months ago

Overview

When you deposit ASTR tokens to a new address for the first time, you may notice the final balance is slightly less than the deposited amount. Specifically, 0.0000000000001 ASTR is deducted from the first deposit to each new address.

For example:

- You deposit: 200 ASTR
- Final balance: 199.999999999999 ASTR
- Difference: 0.0000000000001 ASTR

This is not a bug or rounding error. This is how the Astar Network blockchain is designed to work.

Why does this happen?

The Astar Network charges a small storage rent fee when new on-chain data is created. This fee covers the cost of storing your address information permanently across the entire Astar Network blockchain.

What triggers this fee?

The first deposit only. When ASTR tokens are first sent to a new EVM address, the blockchain needs to:

1. Create a mapping between your EVM address (like MetaMask) and a native ASTR address
2. Store this mapping permanently on-chain
3. Charge a small storage rent fee to cover this data storage

After the first deposit, no additional storage rent is charged because the address mapping already exists.

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Astar's transaction fees include a `rent_fee` component for storing data on-chain: "Storage Rent Fees: Certain on-chain actions incur rent deposits that are refunded when storage is removed. Actions triggering rent include identity creation, multisigs, proxies, and smart contract interactions."

First transactions create storage entries

This behavior is documented in [Astar's Transaction Fees Guide](#): "The first call will usually be more expensive than the following ones because the first call will create a lot of new entries for the user. From the second call, it should be way cheaper (or free) because it will just update those items."

EVM address mapping system

This behavior is documented in [Astar's Account Unification Guide](#).

ASTR uses two types of addresses simultaneously:

- **EVM addresses (H160)**: Used by wallets like MetaMask.
- **Native ASTAR addresses (SS58)**: Used internally by the blockchain.

Every EVM address automatically gets a corresponding native address. This mapping requires on-chain storage, which is why the storage rent fee is charged.

For more information, refer to the explanation in [the EVM-Substrate Guide](#).

How to handle this in your operations

For balance reconciliation

When calculating expected balances for ASTR accounts:

1. **Track first deposits**: Count how many addresses received their first deposit.
2. **Account for storage rent**: Subtract 0.000000000001 ASTR per first deposit.

You can calculate the expected balance using this formula:

$$\text{Expected Balance} = \text{Total Deposits} - (\text{Number of First Deposits} \times 0.000000000001)$$

For withdrawals

To avoid withdrawal failures:

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FAQ

Is this normal Astar behavior?

Yes:

- Astar itself charges the storage rent
- It happens automatically when new address data is created
- This is documented in Astar's official fee model

Why is it not visible in transaction records?

The storage rent fee is deducted internally by the blockchain after the transaction is confirmed. It doesn't appear as a separate transaction because it's part of the blockchain's internal accounting system.

Is the Fireblocks balance accurate?

Yes:

- **Transaction amount:** Fireblocks correctly records the exact amount sent (e.g., 200 ASTR).
- **Final balance:** Fireblocks correctly queries the actual balance from the blockchain after storage rent is deducted (e.g., 199.999999999999 ASTR).

The difference exists on the Astar Network blockchain, not in Fireblocks.

Summary

- **What:** First deposits to ASTR addresses are reduced by 0.000000000001 ASTR.
- **Why:** Astar charges storage rent to create and store your address mapping.
- **Is this normal?:** Yes, this is documented Astar blockchain behavior.
- **Can it be avoided?:** No, this is part of how Astar works.
- **What should you do?:** Account for this small deduction in your balance reconciliation and withdrawal processes.

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- **Contact Astar Network Support:** [Astar Forum](#) or [GitHub](#)
- **Contact Fireblocks Support:** For questions about how Fireblocks handles ASTR transactions.

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